

The circle heat source in the original arithmetic quotient metrics

Split-Zero research programme

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The earlier circle heat construction in [1] has an exact dyadic map to the theta source of the current programme. Equations D1–D3 of the corpus receiver [2] prove the source identity and its Gram error. Here we extend that construction to the full finite relation minimum, the original additive tensor source, its complete primary polynomial and every fixed restriction or quotient of the attained metric. All coordinate maps and the finite error constants are specified below. The truncations need not span the same finite relation subspace; the comparison is proved on the full coefficient space before taking either minimum. The original polynomial and both regions of every source integral are retained throughout.

1 The source, the physical measure and exact division

Retain $x > 0$, $D = -x\partial_x$, $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$ and

$$\begin{aligned}\phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), & f_j &= D^j \Theta\phi_*, \\ V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^a |D^b F(x)| < \infty \text{ for all } a \in \mathbb{R}, b \geq 0\}, \\ \mathcal{M}F(w) &= \int_0^\infty F(x)x^{w-1} dx, & g(w) &= 2\xi(w) = w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w).\end{aligned}\tag{HM1}$$

The completed-zeta convention is Apostol's convention as recorded in [4]. Poisson summation gives $\Theta V \subset \mathcal{B}$: the theta sum of an even Schwartz function has near zero the two terms $x^{-1} \int \phi - \phi(0)$ and a remainder decreasing faster than every power, and those two terms vanish on V . The same argument applies after every D derivative. Since $D(D-1)e^{-\pi x^2} = \phi_*$ and D preserves both zero conditions on V , every f_j belongs to $\mathcal{B}$. For $\Re w > 1$, absolute integration of the theta series and integration by parts give

$$\mathcal{M}f_j(w) = w^j g(w).\tag{HM2}$$

Rapid decrease at both ends extends this identity to the whole plane.

Mellin Parseval in the physical measure is

$$\|F\|_{\mathcal{H}}^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{M}F(1/2 + iy)|^2 dy.\tag{HM3}$$

For precision, put $x = e^u$ and $\tilde{F}(u) = e^{u/2}F(e^u)$. Its $L^2(du)$ norm equals $\|F\|_{\mathcal{H}}$ and its Fourier transform with positive exponential is $(2\pi)^{-1/2}\mathcal{M}F(1/2 + iy)$. Fourier Plancherel proves (HM3), including complex functions and polarized inner products. This is the convention of [3, §1.14(iv)].

Let Z be any nonempty finite set of actual zeros of g , with their full multiplicities m_ρ , and put

$$h(w) = \prod_{\rho \in Z} (w - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \prod_{\rho \in Z} \mathbb{C}[w]/(w - \rho)^{m_\rho}.\tag{HM4}$$

The map j_h takes every divided Taylor coefficient through order $m_\rho - 1$ in the literal coordinate $w - \rho$. In particular $j_h(g/h)$ is a unit in E_h ; no root difference or Taylor coefficient of this unit is omitted.

Lemma 1 (Division in the original source space). *For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define*

$$(S_\rho F)(x) = x^{-\rho} \int_x^\infty F(t) t^{\rho-1} dt.$$

Then $S_\rho F \in \mathcal{B}$, $(D - \rho)S_\rho F = F$, and $\mathcal{M}(S_\rho F)(w) = \mathcal{M}F(w)/(w - \rho)$ with its removable value at ρ . Consequently there is a unique $a_h \in \mathcal{B}$ with $\mathcal{M}a_h = g/h$.

Proof. At infinity rapid decrease of F bounds the integral and each derivative by arbitrarily high negative powers. At zero the vanishing total integral rewrites it as $-x^{-\rho} \int_0^x F(t) t^{\rho-1} dt$, giving arbitrarily high positive powers. Differentiation gives $D(S_\rho F) = \rho S_\rho F + F$ and proves the same estimates for all derivatives. Mellin integration by parts gives the transform formula for $w \neq \rho$, and both sides are entire after removal. Starting from f_0 , divide successively by the complete multiset of roots of h . Each required Mellin zero remains until its corresponding factor is removed. The result has transform g/h . Mellin injectivity, which follows from (HM3), proves uniqueness and independence of the chosen division order. $\square$

For $u \in E_h$, let $r_h(u) \in \mathcal{P}_{d-1}$ be the unique polynomial with

$$j_h r_h(u) = j_h(h/g)u, \quad R_h u = r_h(u)(D)a_h. \quad (\text{HM5})$$

Here h/g is only used as its holomorphic germ at the specified zeros; the notation means the inverse of $j_h(g/h)$. Then $\mathcal{M}R_h u = gr_h(u)/h$ and $j_h \mathcal{M}R_h u = u$ exactly.

2 The full finite prequotient and the heat replacement

Fix $N \geq d - 1$. Write $\mathcal{P}_n = \mathbb{C}[w]_{\leq n}$ and $\mathcal{P}_{-1} = 0$. The following are mutually inverse coordinate maps:

$$\begin{aligned} L_N : E_h \oplus \mathcal{P}_{N-d} &\longrightarrow \mathcal{P}_N, & (u, a) &\longmapsto r_h(u) + ha, \\ L_N^{-1} : P &\longmapsto \left(j_h((g/h)P), \frac{P - r_h(j_h((g/h)P))}{h} \right). \end{aligned} \quad (\text{HM6})$$

Indeed the first component of the inverse recovers u since $j_h g = 0$; the second is a polynomial since its numerator has all the required Taylor zeros. Both compositions are the identity, including when the relation summand is zero. Use the original jet coordinates and the ordinary power coefficients for every matrix below. Define

$$\begin{aligned} F_{N-d}a &= a(D)f_0, \quad \mathcal{R}_N = [R_h, F_{N-d}], \quad \mathcal{P}_N = [0, F_{N-d}], \quad \mathbb{G}_N = \mathcal{R}_N^* \mathcal{R}_N > 0, \\ \mathcal{M}(\mathcal{R}_N L_N^{-1}P) &= \frac{g}{h}P, \quad \left\| \mathcal{R}_N L_N^{-1}P \right\|_{\mathcal{H}}^2 = \int_{\mathbb{R}} |P(1/2 + iy)|^2 w_h(y) dy, \\ w_h(y) &= \frac{1}{2\pi} |(g/h)(1/2 + iy)|^2. \end{aligned} \quad (\text{HM7})$$

This is the original complete one-factor polynomial source. Positivity follows because the nonzero entire function g/h cannot vanish on an interval, so a zero source norm forces $P = 0$ and then $(u, a) = 0$.

On the circle $z \in \mathbb{R}/\mathbb{Z}$, keep $h_0 = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}$ and $\mathcal{N} = -(4\pi^2)^{-1} \partial_z^2$. The old construction [1] is $C(t) = \partial_z(e^{-t\mathcal{N}}h_0)(0)$. Its Fourier coefficients give $C(t) = 4 \sum_{n \text{ odd}} e^{-tn^2}$, where the sum

is over all odd integers. To avoid any dependence on a formal infinite rearrangement, for every integer $J \geq 0$ put $s = 2^{J+1}$ and use the finite identity

$$\frac{1}{4} \sum_{j=0}^J C(\pi 4^j x^2) = \vartheta(x^2) - \vartheta(s^2 x^2), \quad \vartheta(v) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 v}. \quad (\text{HM8})$$

This follows by separating odd from even integers at each step. Applying $D^{J+1}(D-1)$ (with the derivative index j independent of the summation index in (HM8)) gives the exact source replacement

$$\begin{aligned} T_s F(x) &= F(sx), & T_s^* T_s &= s^{-1} I, \\ f_j^{[J]} &= (I - T_s) f_j = \frac{1}{4} D^{J+1} (D-1) \sum_{\ell=0}^J C(\pi 4^\ell x^2), \\ \mathcal{R}_N^{[J]} &= \mathcal{R}_N - T_s \mathcal{P}_N, & \mathbb{G}_N^{[J]} &= (\mathcal{R}_N^{[J]})^* \mathcal{R}_N^{[J]}. \end{aligned} \quad (\text{HM9})$$

All columns of $T_s \mathcal{P}_N$ lie in ΘV . Therefore the two full column maps have the same image in $\mathcal{B}/\Theta V$ and the same full jets at every actual zero. This equality of quotient classes alone makes no assertion about either finite metric; those metrics are compared next.

3 An explicit comparison before taking any minimum

When $N \geq d$, partition the full original Gram as

$$\mathbb{G}_N = \begin{pmatrix} A & C & C^* & S \end{pmatrix}, \quad H = A - C S^{-1} C^* > 0, \quad c_N^2 = 1 + \left\| H^{-1/2} C S^{-1/2} \right\|^2. \quad (\text{HM10})$$

All norms on finite coordinate spaces here are Euclidean operator norms; the positive square roots are those of the actual matrices. When $N = d - 1$ there is no relation block and define $c_N = 0$.

Theorem 2 (Full finite heat comparison). *For every $N \geq d - 1$ and every $J \geq 0$, let $\delta_{N,J} = 2^{-(J+1)/2} c_N$. For $\delta_{N,J} < 1$,*

$$(1 - \delta_{N,J})^2 \mathbb{G}_N \preceq \mathbb{G}_N^{[J]} \preceq (1 + \delta_{N,J})^2 \mathbb{G}_N. \quad (\text{HM11})$$

In particular the actual minima over all relation coefficients obey

$$(1 - \delta_{N,J})^2 H \preceq H^{[J]} \preceq (1 + \delta_{N,J})^2 H, \quad u^* H^{[J]} u = \min_a \|R_h u + (I - T_s) F_{N-d} a\|_{\mathcal{H}}^2. \quad (\text{HM12})$$

The matrices in (HM10) are sufficient to choose a finite J for every specified positive error tolerance; no growth assumption on c_N is used.

Proof. For the nonempty relation block, $\mathcal{P}_N^* \mathcal{P}_N = \text{diag}(0, S)$. The nonzero eigenvalues of $\mathbb{G}_N^{-1/2} \mathcal{P}_N^* \mathcal{P}_N \mathbb{G}_N^{-1/2}$ are those of the bottom principal block of $\mathbb{G}_N^{-1}$ conjugated by $S^{1/2}$. Direct block elimination gives that block as $S^{-1} + S^{-1} C^* H^{-1} C S^{-1}$. Its conjugate is $I + S^{-1/2} C^* H^{-1} C S^{-1/2}$, whose largest eigenvalue is c_N^2 . This is also obtained by completing the square

$$\|\mathcal{R}_N(u, a)\|^2 = u^* H u + (a + S^{-1} C^* u)^* S (a + S^{-1} C^* u).$$

Thus, for every coefficient vector v ,

$$\|T_s \mathcal{P}_N v\|_{\mathcal{H}} \leq s^{-1/2} c_N \|\mathcal{R}_N v\|_{\mathcal{H}}. \quad (\text{HM13})$$

The two triangle inequalities for $\mathcal{R}_N^{[J]} v = \mathcal{R}_N v - T_s \mathcal{P}_N v$ prove (HM11). The zero-relation case is an equality. To prove (HM12), use (HM11) on every vector (u, a) and take the infimum over the same coefficient a on each side. Positive definiteness ensures the minima are attained. This proves the comparison even though $F_{N-d} \mathcal{P}_{N-d}$ and $(I - T_s) F_{N-d} \mathcal{P}_{N-d}$ need not be the same subspace of $\mathcal{H}$. $\square$

For reference, the exact correction, with both cross terms retained, is

$$\mathbb{G}_N^{[J]} - \mathbb{G}_N = -\mathcal{R}_N^* T_s \mathcal{P}_N - \mathcal{P}_N^* T_s^* \mathcal{R}_N + s^{-1} \mathcal{P}_N^* \mathcal{P}_N. \quad (\text{HM14})$$

Completing a positive quadratic form and its minimum-norm lift is the classical Schur-complement calculation of [5, Appendix C.4.1]; the preceding proof specifies its matrices in this arithmetic source.

4 The exact additive tensor source and its full primary quotient

Let $k \geq 1$ and retain each separate Mellin coordinate $w_j = 1/2 + iy_j$. On the original S -power coefficients define the literal multinomial map

$$\begin{aligned} \mathcal{S}_{k,N} : \mathbb{C}[S]_{\leq N} &\longrightarrow \mathcal{P}_N^{\otimes k}, \\ S^n &\longmapsto \sum_{n_1 + \dots + n_k = n} \frac{n!}{n_1! \dots n_k!} w_1^{n_1} \dots w_k^{n_k}, \\ \mathcal{C}_{k,N} &= (L_N^{-1})^{\otimes k} \mathcal{S}_{k,N}, \\ \mathcal{A}_{k,N} &= \mathcal{R}_N^{\otimes k} \mathcal{C}_{k,N}, \quad \mathcal{A}_{k,N}^{[J]} = (\mathcal{R}_N^{[J]})^{\otimes k} \mathcal{C}_{k,N}. \end{aligned} \quad (\text{HM15})$$

These maps target $\mathcal{H}^{\otimes k} = L^2((\mathbb{R}_{>0})^k, dx_1 \dots dx_k)$. In particular no new product, root polynomial or coefficient normalization is inserted. The multiple Mellin transform of the first map is exactly

$$\mathcal{M}^{\otimes k}(\mathcal{A}_{k,N} P)(w_1, \dots, w_k) = P(w_1 + \dots + w_k) \prod_{j=1}^k (g/h)(w_j). \quad (\text{HM16})$$

Tensor Parseval and Tonelli, followed by the definition of convolution, therefore prove

$$\begin{aligned} \|\mathcal{A}_{k,N} P\|^2 &= \int_{\mathbb{R}^k} |P(k/2 + i \sum_j y_j)|^2 \prod_j w_h(y_j) dy_j \\ &= \int_{\mathbb{R}} |P(k/2 + iy)|^2 w_h^{*k}(y) dy. \end{aligned} \quad (\text{HM17})$$

The entire real line is retained. All moments are finite by (HM7) and the rapid vertical decrease of Mellin transforms of $\mathcal{B}$. The form in (HM17) is the original tensor source of [6]; denote its power-basis matrix by $\mathcal{H}_{k,N} > 0$ and the corresponding heat matrix by $\mathcal{H}_{k,N}^{[J]}$.

The finite tensor quotient class is preserved as well. Indeed, expanding the difference of the tensor maps in (HM15) by replacing factors one at a time expresses it as a sum of tensors with one factor in ΘV . Consequently both maps induce the same map into the algebraic tensor quotient $\mathcal{B}^{\otimes k} / \sum_i \mathcal{B}^{\otimes(i-1)} \otimes \Theta V \otimes \mathcal{B}^{\otimes(k-i)}$. No identification with a completed tensor quotient is needed for the following finite matrix comparison.

Theorem 3 (The original tensor minima receive the heat error). *For $\delta = \delta_{N,J} < 1$ one has*

$$(1 - \delta)^{2k} \mathcal{H}_{k,N} \preceq \mathcal{H}_{k,N}^{[J]} \preceq (1 + \delta)^{2k} \mathcal{H}_{k,N}. \quad (\text{HM18})$$

Let $\chi(S)$ be the original complete primary polynomial of the packet, $q = \deg \chi$, and take $N \geq \max(d-1, q-1)$. Keep the remainder map $J_{\chi,N} : \mathbb{C}[S]_{\leq N} \rightarrow E_\chi = \mathbb{C}[S]/(\chi)$ in its original coordinates. The two actual attained quotient forms satisfy

$$\begin{aligned} G_{k,N} &= (J_{\chi,N} \mathcal{H}_{k,N}^{-1} J_{\chi,N}^*)^{-1}, \quad G_{k,N}^{[J]} = (J_{\chi,N} (\mathcal{H}_{k,N}^{[J]})^{-1} J_{\chi,N}^*)^{-1}, \\ (1 - \delta)^{2k} G_{k,N} &\preceq G_{k,N}^{[J]} \preceq (1 + \delta)^{2k} G_{k,N}. \end{aligned} \quad (\text{HM19})$$

All original multiplicities, roots and the complete relation space $\chi \mathbb{C}[S]_{\leq N-q}$ remain unchanged in this comparison.

Proof. Conjugating (HM11) by $\mathbb{G}_N^{-1/2}$ puts every eigenvalue between $(1 - \delta)^2$ and $(1 + \delta)^2$. The eigenvalues of its k -fold tensor power are the products of k such eigenvalues. Returning to the original coordinate forms gives the corresponding bounds on $(\mathbb{G}_N^{[J]})^{\otimes k}$. Pull them back through the exact map $\mathcal{C}_{k,N}$ to obtain (HM18).

For completeness, for any positive matrix M and any surjection J , the minimum of p^*Mp subject to $Jp = v$ is $v^*(JM^{-1}J^*)^{-1}v$. Its unique minimum-norm representative is $p_* = M^{-1}J^*(JM^{-1}J^*)^{-1}v$; every other representative is $p_* + z$ with $Jz = 0$ and $z^*Mp_* = 0$. This proves both formulas in (HM19). Taking the infimum in (HM18) over the identical affine fibre $J_{\chi,N}p = v$ gives its comparison. Its kernel is exactly the specified full polynomial ideal by Euclidean division by χ . $\square$

For the quartet used in the current programme, this theorem retains literally

$$\begin{aligned} \rho &= \tfrac{1}{2} + \delta_\rho + i\gamma, \quad 0 < \delta_\rho < \tfrac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \\ k &\equiv 1 \pmod{4}, \quad k \geq 5, \quad h(w) = \prod_{\sigma \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (w - \sigma)^{m_0}, \\ e &= 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \\ \chi(S) &= \prod_{a,b=0}^k [S - k/2 - (2a - k)\delta_\rho - i(2b - k)\gamma]^e. \end{aligned} \tag{HM20}$$

In particular the dyadic s is a heat truncation parameter and is not identified with k, q , a period, a root displacement or the physical coordinate $S = k/2 + iy$.

5 Restrictions, further quotients and an attained finite budget

Let M be any one of the forms just compared, and let $M^{[J]}$ be its heat counterpart in the same coefficient space. For every fixed injective frame I and fixed surjection Λ , define

$$M_I = I^*MI, \quad Q_\Lambda = (\Lambda M^{-1}\Lambda^*)^{-1}. \tag{HM21}$$

Multiplication by I preserves both inequalities in (HM18)–(HM19). Taking the infimum over $\Lambda v = u$ preserves them by the proof of (HM19). Induction gives the same constants for every finite sequence of these fixed restrictions and quotients, including the original conductor kernel and attained observation when their original coefficient maps are used. This statement fixes the maps before changing the metric; it does not substitute metric-dependent orthonormal frames for them.

If the resulting form has rank $r > 0$, each eigenvalue relative to its original form lies between $(1 - \delta)^{2k}$ and $(1 + \delta)^{2k}$. Hence its determinant in the same coordinate frame obeys

$$\left| \log \frac{\det M_X^{[J]}}{\det M_X} \right| \leq 2kr[-\log(1 - \delta)]. \tag{HM22}$$

Rank zero contributes determinant 1 and error zero. For a finite specified return $\mathcal{L} = \sum_\ell a_\ell \log \det M_{X_\ell}$, let $B = \sum_\ell |a_\ell| r_\ell$. If its cutoff family is finite, use $c_* = \max_N c_N$ over exactly those original prequotients, the same J for the whole family, and $\delta = 2^{-(J+1)/2} c_*$. Then

$$|\mathcal{L}^{[J]} - \mathcal{L}| \leq 2kB[-\log(1 - \delta)]. \tag{HM23}$$

This includes a full original cutoff window with its literal endpoint weights, once their ranks and coefficients are inserted into B . It does not discard a rank factor from an absolute determinant estimate.

Here is a finite choice that actually attains every prescribed error $\varepsilon > 0$. If $B = 0$ or $c_* = 0$, the error is zero. Otherwise put

$$\delta_* = \min \left\{ \frac{1}{4}, \frac{\varepsilon}{4kB} \right\}, \quad J = \max \{0, \lceil 2 \log_2(c_*/\delta_*) \rceil - 1\}. \quad (\text{HM24})$$

This gives $\delta \leq \delta_* \leq 1/4$. The integral identity $-\log(1 - \delta) = \int_0^\delta (1 - t)^{-1} dt$ gives $-\log(1 - \delta) \leq 4\delta/3$, so (HM23) is at most ε (indeed at most $2\varepsilon/3$). All matrices defining c_* are the full original finite integral Grams in (HM7), with the same arithmetic unit and the two cross blocks in (HM10). Thus (HM24) proves finite convergence of these geometric heat sources to every specified original metric return. It does not assert a bound on the growth of c_N , and in particular makes no assertion that this choice of J is logarithmic in q .

6 The actual nilpotent graded metrics and their singular values

The recovered jet-ideal chain also has the following precise receiver. For the full original sum polynomial χ , write its distinct roots as κ with their original multiplicities e_κ . Its confluent matrix in the original power basis is

$$V_{(\kappa, j), n} = \binom{n}{j} \kappa^{n-j} \quad (0 \leq n < q, \ 0 \leq j < e_\kappa),$$

with zero entries when $n < j$. Let N_κ multiply by $t = S - \kappa$ on $\mathbb{C}[t]/t^{e_\kappa}$ and put $N = V^{-1}(\bigoplus N_\kappa)V$. Define the finite increasing filtration

$$M_a E_\chi = V^{-1} \bigoplus_{\kappa} \text{span}\{t^j : e_\kappa - 1 - 2j \leq a\}. \quad (\text{HM25})$$

This is the explicit single-Jordan-block construction in Deligne [7, (1.6.1), (1.6.7)]. Indeed N lowers the displayed weight by two. For $a \geq 0$, a nonzero source grade at κ occurs precisely when $a \leq e_\kappa - 1$ and $a \equiv e_\kappa - 1 \pmod{2}$, and its vector is $t^{(e_\kappa - 1 - a)/2}$. Multiplication by N^a sends it with coefficient one to $t^{(e_\kappa - 1 + a)/2}$. Thus the exact induced map $T_a : \text{Gr}_a^M \rightarrow \text{Gr}_{-a}^M$ is an isomorphism on the stated original coordinate vectors. This proves the nilpotent identities directly for this packet.

Order the columns $V^{-1}e_{\kappa, j}$ by increasing weight, with the same fixed order within ties, and call the resulting matrix C_M . For $\mathcal{B} = C_M^* G_{k, N} C_M$, let $<$ mean all weights below a . The actual metric on M_a/M_{a-1} and its heat counterpart are

$$\mathcal{G}_a = \mathcal{B}_{aa} - \mathcal{B}_{a<} \mathcal{B}_{<<}^{-1} \mathcal{B}_{<a}, \quad \mathcal{G}_a^{[J]} = \mathcal{B}_{aa}^{[J]} - \mathcal{B}_{a<}^{[J]} (\mathcal{B}_{<<}^{[J]})^{-1} \mathcal{B}_{<a}^{[J]}. \quad (\text{HM26})$$

For the first grade the second term is absent. These are precisely restrictions followed by the quotient minima in (HM21): minimizing $(v, u)^* \mathcal{B}(v, u)$ over v gives the unique lift $v = -\mathcal{B}_{<<}^{-1} \mathcal{B}_{<a} u$ and (HM26). Consequently

$$l \mathcal{G}_a \preceq \mathcal{G}_a^{[J]} \preceq u \mathcal{G}_a, \quad l = (1 - \delta)^{2k}, \quad u = (1 + \delta)^{2k}. \quad (\text{HM27})$$

In particular no cross block between different primary states or different nilpotent grades is suppressed. The frame here is a specified complex coefficient frame, without an assertion that it is an automorphism of a free split-semiring module.

Let $\lambda_{a, i} > 0$ be the ordered squared singular values of T_a in the actual two graded metrics, and use $\lambda_{a, i}^{[J]}$ for the heat metrics. Explicitly the former are the eigenvalues of $\mathcal{G}_a^{-1/2} T_a^* \mathcal{G}_{-a} T_a \mathcal{G}_a^{-1/2}$. Equation (HM27) bounds the Rayleigh quotient for every nonzero vector between l/u and u/l

times its original value. Applying the finite min–max formula (minimizing the maximum over each subspace of the appropriate dimension) gives, for each ordered eigenvalue,

$$\frac{l}{u}\lambda_{a,i} \leq \lambda_{a,i}^{[J]} \leq \frac{u}{l}\lambda_{a,i}, \quad \left| \log \frac{\lambda_{a,i}^{[J]}}{\lambda_{a,i}} \right| \leq 2k \log \frac{1+\delta}{1-\delta}. \quad (\text{HM28})$$

This bounds the singular values of the actual induced maps, as well as the separate graded determinants. Iterated block elimination also gives

$$\det G_{k,N} = |\det V|^2 \prod_a \det \mathcal{G}_a, \quad \det G_{k,N}^{[J]} = |\det V|^2 \prod_a \det \mathcal{G}_a^{[J]}. \quad (\text{HM29})$$

Indeed $|\det C_M| = |\det V|^{-1}$, and each elimination factor in (HM26) is the actual Schur factor of the same full Gram. These constructions implement Deligne’s nilpotent filtration and its metric receivers in the original sum packet. They specify the graded operators whose estimates a mixed-cohomology argument can use. No arithmetic Frobenius operator or weight-purity theorem is inferred from the integers in (HM25).

On the literal quartet sum (HM20), all primary multiplicities are e . Each nonzero grade therefore has dimension $(k+1)^2$, and for $a \geq 0$ with $a = e-1 \pmod{2}$, $a \leq e-1$, the same ordering of roots makes T_a the identity between those two different graded spaces. Its squared singular values are still the eigenvalues of $\mathcal{G}_a^{-1/2} \mathcal{G}_{-a} \mathcal{G}_a^{-1/2}$; the two metrics need not coincide. In particular, for the simple-quartet case $m_0 = 1$, one has $e = 1$, and the exact specialization is

$$M_{-1} = 0, \quad M_0 = E_\chi, \quad N = 0, \quad \mathcal{G}_0 = V^{-*} G_{k,N} V^{-1}, \quad T_0 = I, \quad \lambda_{0,i} = 1. \quad (\text{HM30})$$

Thus this specialization retains the entire q -dimensional arithmetic metric in its sole grade. The nilpotent comparison gives no additional separation of its distinct root states or evaluation of their observed angles. Their full original metric is carried by (HM19) and (HM30); those cross-primary calculations remain necessary.

7 Consequences for the current calculation

The source identity from the earlier paper has now been carried through the exact maps

$$\text{circle heat} \longrightarrow (I - T_s) F_{N-d} \longrightarrow \mathcal{R}_N^{[J]} \longrightarrow \mathcal{A}_{k,N}^{[J]} \longrightarrow G_{k,N}^{[J]} \longrightarrow (G_{k,N}^{[J]})_X.$$

Equations (HM6), (HM15), (HM19) and (HM21) define each coefficient map; (HM11), (HM18), (HM22) and (HM24) supply its full quantitative control. In particular the original kernel restriction in FW51–52 of [6] and its complementary observation quotient both receive this comparison. For either entire-window determinant return, one may choose (HM24) with its full B and any $\varepsilon > 0$, such as $\varepsilon = 1$. This transfers that return with its complete signs, weights and affine minima. It supplies no missing arithmetic allocation coefficient: it constructs a convergent geometric realization of the original metric in which those coefficients are to be calculated. The exact theta-class identity precedes the comparison, so closeness of metrics is never substituted for equality of the original class.

References

- [1] Earlier Split-Zero corpus, *Corrected Wick-Rotated Characters, Twisted Spectral Traces, Lorentzian Lifts, and the Periodic Spline-Wavelet Hierarchy*, May 2026, source file `corrected_wick_rotated_spline_wavelet_note.tex`, labels `mod:g0-fourier`, `mod:spline-primitive` and `mod:odd-square`, original source lines 646–774. The circle Fourier construction is used; unrelated physical claims are not imported. The exact source file and its hash accompany this proof.

- [2] Split-Zero corpus recovery, *Exact receivers from earlier Split Zero mathematics*, 19 September 2026, Theorem 1 and Proposition 2, equations D1–D3. The heat source and norm identities are reproduced in full above.
- [3] Ranjan Roy, Frank W. J. Olver, Richard A. Askey, Roderick Wong and William P. Reinhardt, Chapter 1, *NIST Digital Library of Mathematical Functions*, §1.14(iv), equations 1.14.32 and 1.14.38, <https://dlmf.nist.gov/1.14>. Used for Mellin and Parseval conventions; the change to the physical measure dx is shown in (HM3).
- [4] Tom M. Apostol, Chapter 25, *NIST Digital Library of Mathematical Functions*, equation 25.4.4, <https://dlmf.nist.gov/25.4.E4>.
- [5] Stephen Boyd and Lieven Vandenbergh, *Convex Optimization*, Cambridge University Press, 2004, Appendix C.4.1, https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. The original cross-block calculation is proved explicitly in (HM10)–(HM13).
- [6] Split-Zero research programme, *Full-window source products and the retained class: finite quotient comparisons in the original arithmetic metric*, 19 September 2026, FW1–52, particularly FW1–4 and FW37–40, FW51–52. GitHub edition 3c44de632b2c5ead13b23619d646803d2bbffc19.
- [7] Pierre Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l’IHÉS 52 (1980), 137–252, §1.6, especially (1.6.1) and (1.6.7), https://www.numdam.org/item/PMIHES_1980__52__137_0/. The complete D032 typed source accompanies this proof. Equations (HM25)–(HM29) specify and prove its finite packet and metric receiver.